

Cheng-Nan Liu

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SUMMARY

Applied Scientist specializing in **large-scale time-series modeling and low-SNR signal recovery** from distributed sensor networks. Experienced building production ML pipelines—from noisy, multi-TB sensor data to multimodal models—at **Google X** and ML startups. First-author researcher with a track record bridging research prototypes and scalable inference systems.

EDUCATION

University of Utah <i>Ph.D., Geology and Geophysics</i> Dissertation: Seismic Monitoring of Yellowstone Hydrothermal Systems.	Salt Lake City, UT 2021 – May 2026
National Taiwan University <i>M.S., Geoscience</i> Thesis: Low-cost Seismometers and Machine Learning for Onsite Earthquake Early Warning.	Taipei, Taiwan 2017 – 2019
National Cheng Kung University <i>B.S., Earth Science</i>	Tainan, Taiwan 2013 – 2017

WORK EXPERIENCE

X, The Moonshot Factory (prev. Google X) <i>Earth Scientist AI Resident</i>	Mountain View, CA Onsite, Jan. 2026 – May 2026
• Multimodal Contrastive Learning: Aligned heterogeneous sensor time-series with structured metadata using a SIGLIP-based contrastive model, achieving ~99% precision and recall under high-noise conditions. • Recovered clean training signals from extremely noisy large-scale sensor data via self-supervised preprocessing, converting raw unusable data into a labeled corpus at scale. • Built a data-centric SQL pipeline with human-in-the-loop annotation, enabling continuous dataset expansion and iterative retraining while reducing label noise and false positives.	
Lumetec, Inc. <i>Data Scientist Intern</i>	Pasadena, CA Onsite, Jun. 2025 – Aug. 2025
• Sensor Event Fingerprinting: Built an acoustic fingerprinting system using spectral peak hashing to identify unique event signatures from large-scale, noisy sensor streams. • Achieved 70% accuracy on known event types and >90% detection rate on novel, previously unseen events; optimized feature extraction to improve low-SNR robustness.	
Lawrence Berkeley National Laboratory <i>Research Scientist Intern</i>	Berkeley, CA Onsite, May 2024 – Aug. 2024
• Signal Reconstruction from Sparse Inputs: Developed a Swin Transformer model (PyTorch) to reconstruct missing signals from sparse, irregular sensor arrays trained on 50+ TB of data. • Achieved ~0.9 correlation with ground truth up to 20 Hz; accelerated inference 30× (hours → seconds) via an optimized pipeline for large-scale batch processing.	

SKILLS

ML Focus:	Multimodal Learning, Contrastive Representation, Self-Supervised Learning, Signal Reconstruction
Frameworks:	PyTorch, Scikit-learn, Transformers, CNNs, Deep Learning
Data & Systems:	SQL, GCP (BigQuery, Vertex AI), Docker, Git, Cloud Data Pipelines
Domain:	Time-Series Analysis, Signal Processing, Distributed Sensor Networks
Programming:	Python, SQL, MATLAB, Bash